

Department of Physics, FNSPE, CTU in Prague
02YELMA - Electricity and Magnetism
Midterm 2 (A)

15.5.2026 / 16:00 - 17:30

Q1: A long, straight, solid cylindrical conductor of radius R carries a steady total current I in the $+\hat{z}$ direction. The volume current density varies with radial distance s from the axis as $\vec{J}(s) = J_0(s/R)\hat{z}$ when $0 \leq s \leq R$, and $\vec{J}(s) = 0$ when $s > R$. Express J_0 in terms of I and R , and then find the magnetic field $\vec{B}(s)$ for $s < R$ and $s > R$.

Q2: A parallel-plate capacitor with circular plates of radius R is being charged by a steady current I . The plate separation is d , with $d \ll R$, so edge effects may be neglected. Assume vacuum between the plates. At a given instant, the electric field between the plates is uniform and directed along the z -axis $\vec{E}(t) = E(t)\hat{z}$. Using Poynting's theorem, determine the energy stored in the capacitor at a given time. — *Solutions without using Poynting's theorem will not receive credit.*

Q3: The magnetic field component of a monochromatic electromagnetic wave propagating in free space is given by $\vec{B}(z, t) = B_0 \sin(kz - \omega t)\hat{y}$.

- (a) Find the average power transmitted through a circular area of radius a in the plane perpendicular to the direction of propagation.
- (b) If the wave goes through a polarizer oriented at an angle $\pi/4$ with respect to the direction of electric field polarization, find the amplitude of the transmitted electric field.

Q4: A metallic cylinder of radius R and height h carries a total charge Q . It rotates about its symmetry axis $\hat{z}$ with a constant angular velocity ω . Calculate the magnetic dipole moment of the cylinder. — *Assume the charge is uniformly distributed over the curved outer surface.*

Maxwell's equations

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$\oint_S \vec{B} \cdot d\vec{a} = 0$$

$$\oint_C \vec{E} \cdot d\vec{\ell} = -\frac{d}{dt} \int_S \vec{B} \cdot d\vec{a}$$

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 I_{\text{enc}} + \mu_0 \epsilon_0 \frac{d}{dt} \int_S \vec{E} \cdot d\vec{a}$$

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$